

FINPERSONA-BENCH

Synthetic market environment v2

What changed, why, and how we checked it



market generator

agent harness

evaluation layer

Outline

- 1** Shortcomings of the previous environment
what was wrong, in plain terms
- 2** The changes that fix them, one by one, and why
ten changes, each with the reason and a figure from the new generator
- 3** Additions beyond fixing shortcomings
an overview, then the five additions that are not part of a change
- 4** Tests and pilot that confirm they landed
stylized-facts checklist, leakage audit, robustness, a 48-run pilot

All numbers on the test slides are from the final generator, 50 seeds per scenario.

Shortcomings of the previous environment

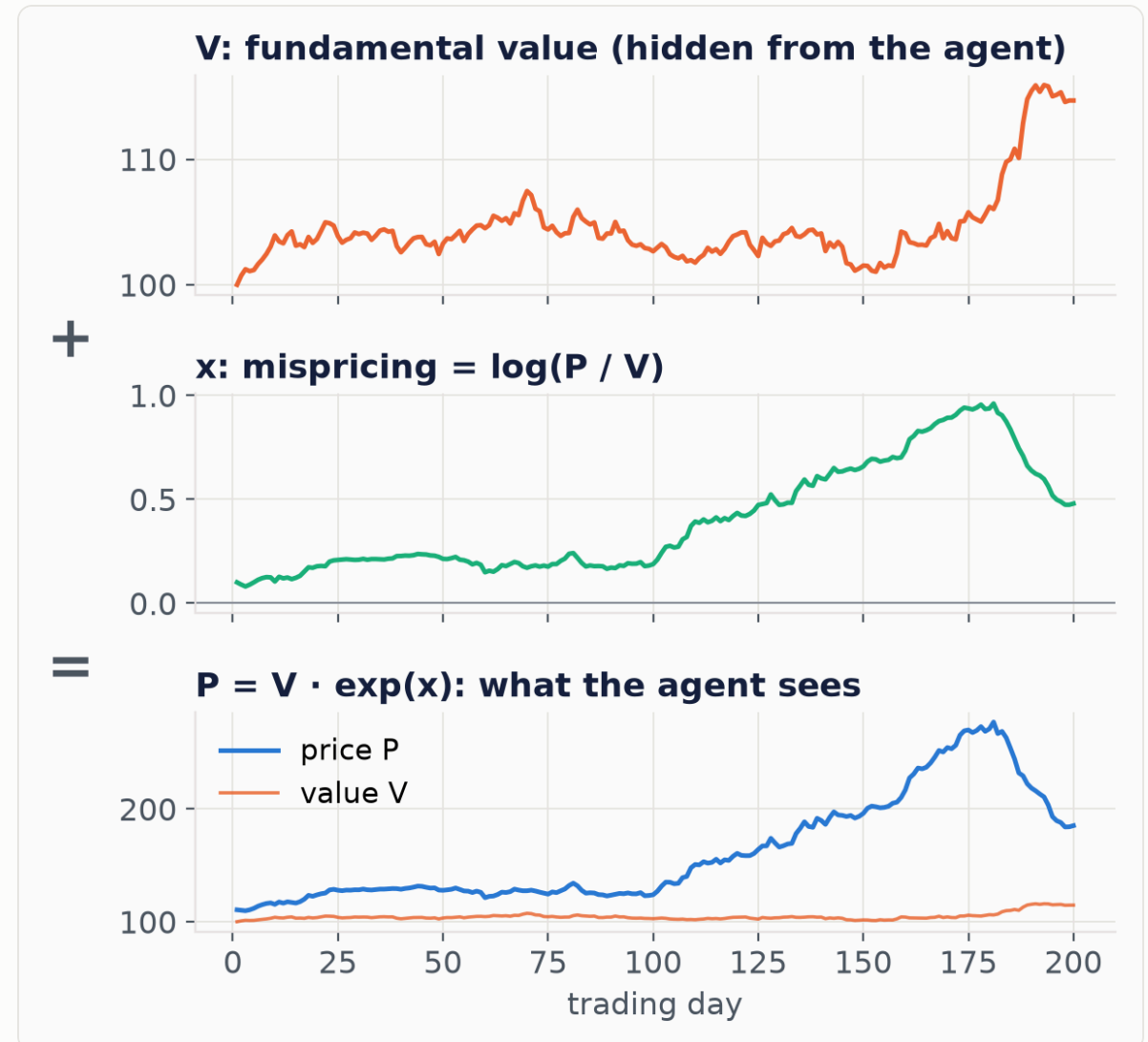
- 1 Valuation fields were tied to the hidden value by a fixed formula
- 2 Phases started on the same day in every run
- 3 Volatility did not cluster and had no fat tails
- 4 The crash-severity setting changed outcomes by under 1 pp
- 5 Volume, sentiment and implied volatility followed the phase label
- 6 Every run started 100 % in cash, and SELL did nothing on day 1
- 7 Target cash levels 1.0 / 0.5 / 0.2 were stipulated
- 8 The prompt showed 10 of the 16 listed fields; no baselines, no cost model

Price = value × mispricing

$\log P = \log V + x$, one stated equation for every scenario

- Value V : an exogenous random walk with fat-tailed shocks, about 6.5 % a year drift and 0.6 % daily volatility
- Mispricing x : generated by a separate process (next slides)
- Price is the product of the two; the agent sees P , never V or x

Why value and price come from separate processes, so the ground truth is explicit and can never be recovered from an accounting identity

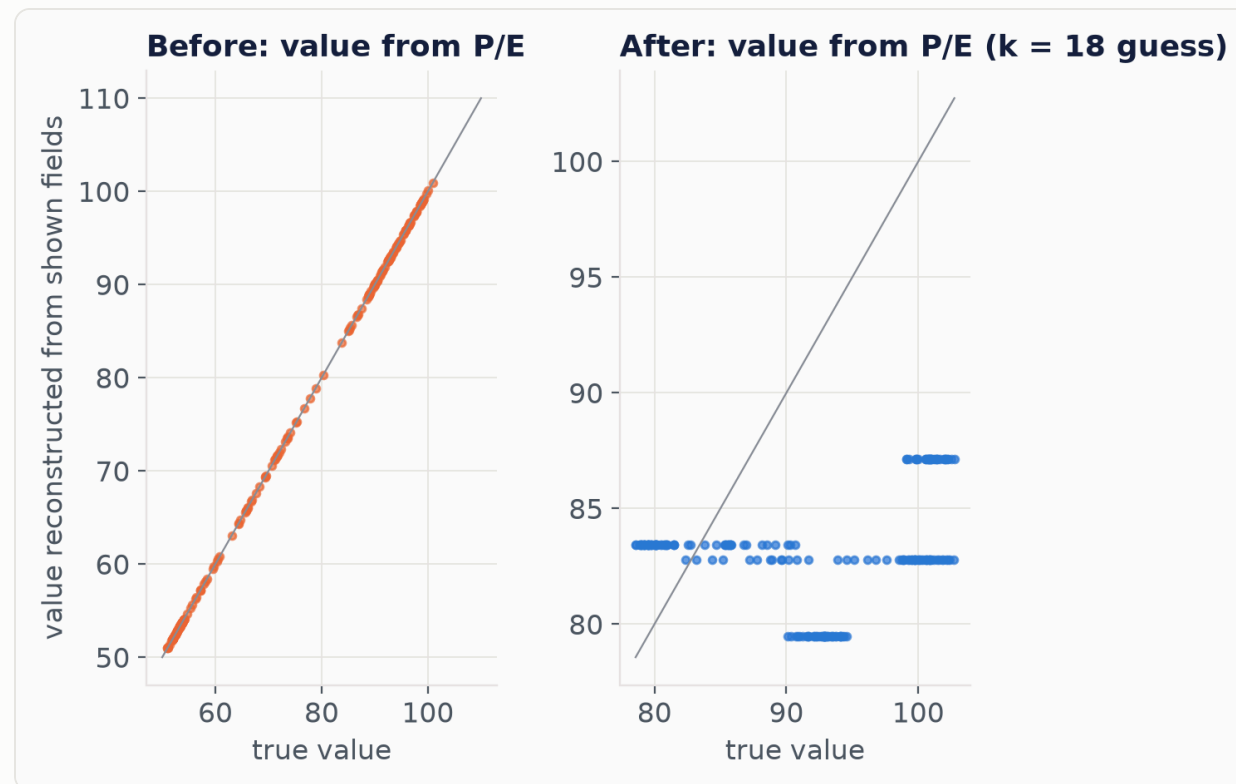


Valuation fields no longer pin down the value

- Earnings: quarterly, reported 25–35 days late, 10 % noise, and a hidden multiple per run (14–22)
- P/E is trailing four quarters; dividends are sticky; the analyst estimate carries a persistent 15 % error

Why the signals still point the right way when mispricing is large, but no formula on the shown fields reproduces V

Best formula on shown fields	Before	After
multiple $\times$ price $\div$ P/E, median error vs V	0.2 %	15 %
analyst fair-value estimate, error vs V	—	22 %

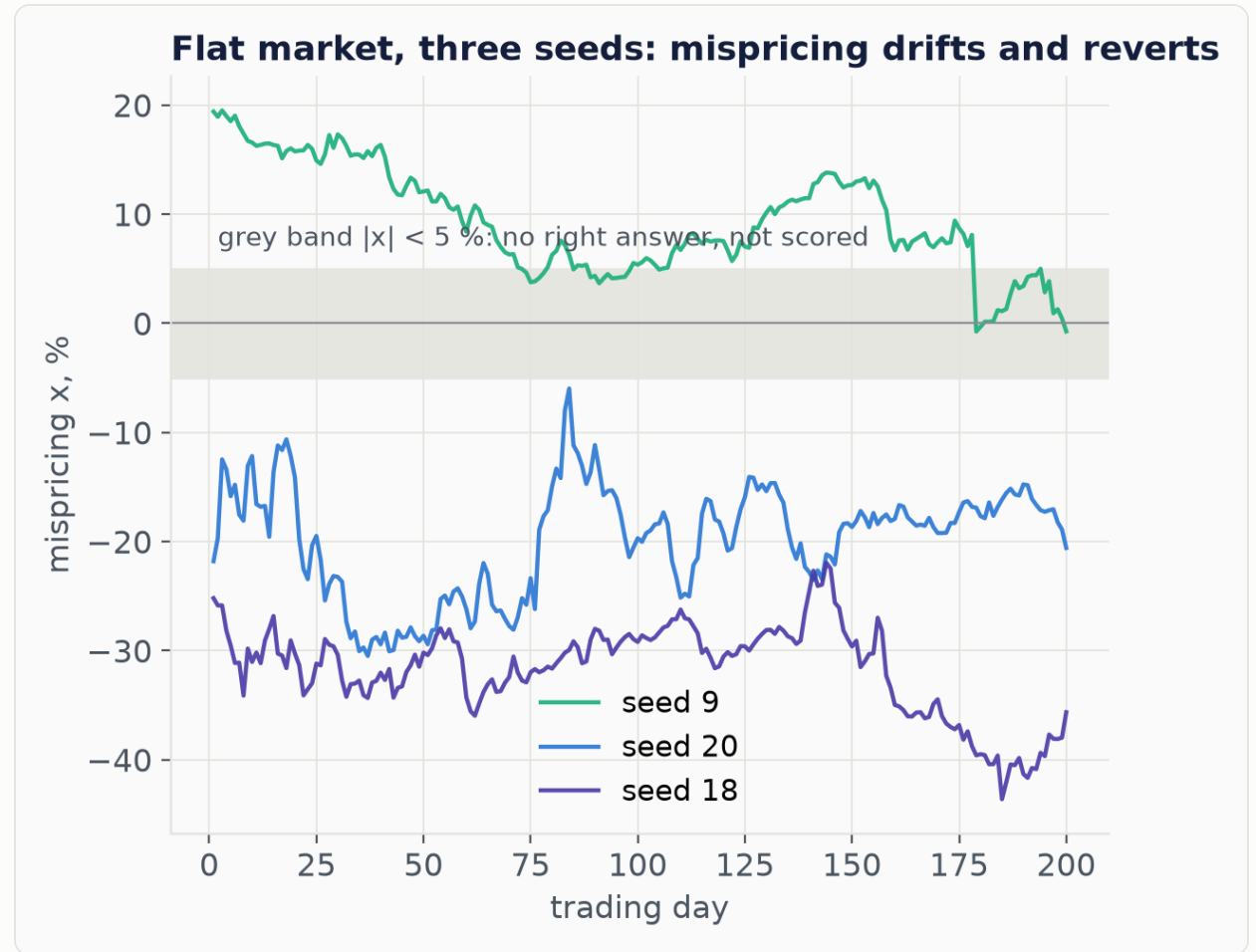


grey line = perfect reconstruction; a hidden multiple and lagged earnings turn the line into a cloud

Mispricing with a published form

- Calm phases: Franke–Westerhoff fundamentalists (pull towards value) and chartists (extrapolate), driven by one innovation
- Persistence: stationary half-life ≈ 150 days (≈ 70 days realised), a stated design value
- Re-estimation on 10 US stocks was attempted; return moments do not identify the persistence, so the published form is used and the published index parameters are kept as a sensitivity

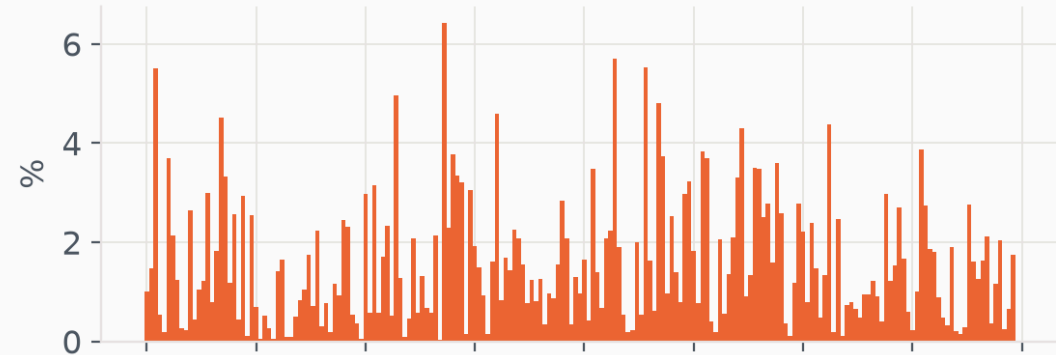
Why mispricing now persists, so even the flat market contains decisions that are right or wrong



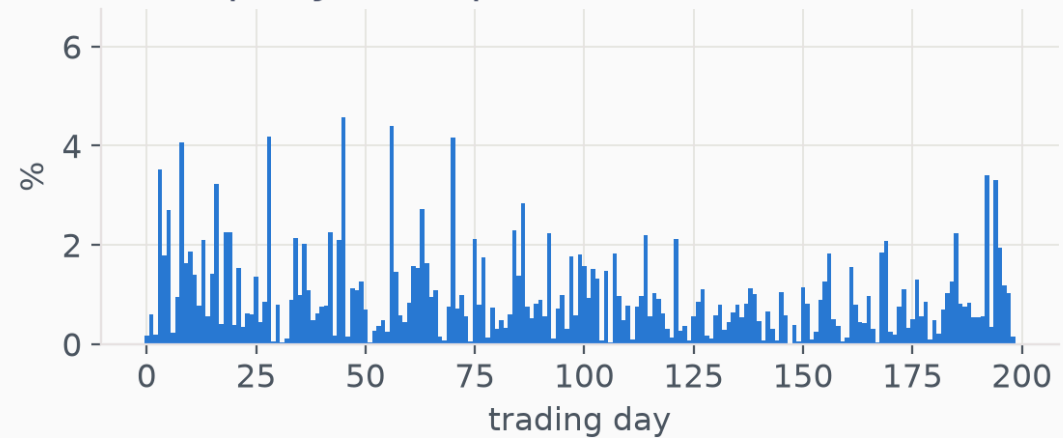
- One innovation process: GJR-GARCH(1,1) with Student-t(5) shocks and rare jumps (1 % of days, $-4\% \pm 3\%$)
- Persistence 0.98; event phases scale the variance (panic $\times 5$, post-top $\times 3$, mania $\times 1.5$)

Why clustering, fat tails and crash asymmetry become properties of the series; the old artificial mean-reversion signal is gone

Before: |daily return| is a flat band, no bursts



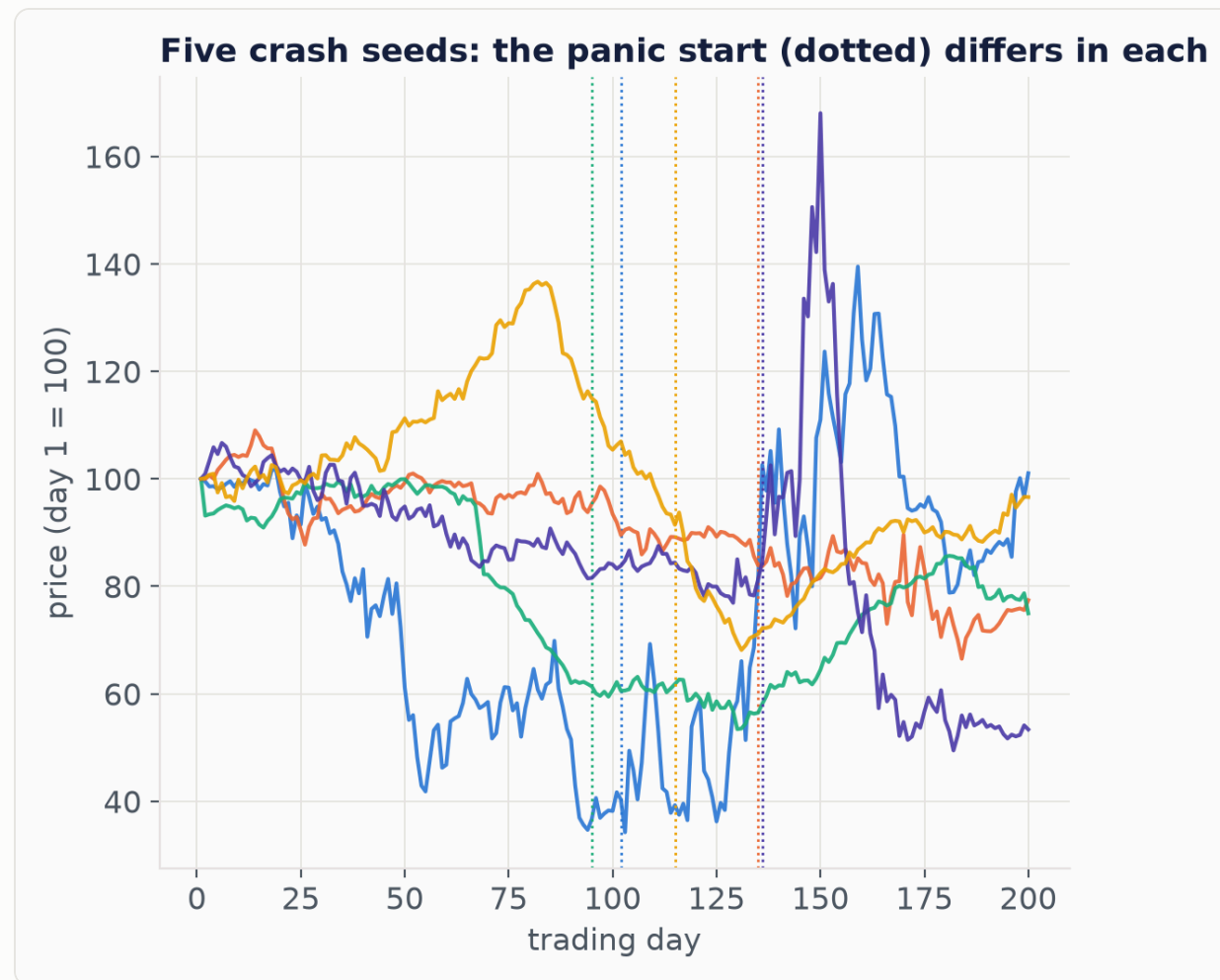
After: |daily return| shows clustered bursts and fat tails



Events scripted in direction, random in timing

- Each run draws its own setup length (25–55 % of the horizon), panic length (15–70 days), crash depth (10–30 %) and discount (55 / 70 / 85 %)
- Bubbles: compounding drift, a probabilistic top (about half of runs top inside the horizon, peak $P/V \approx 2.1$) and a post-top leg
- Sustained-bull control; event-first and phase-free orderings as further controls

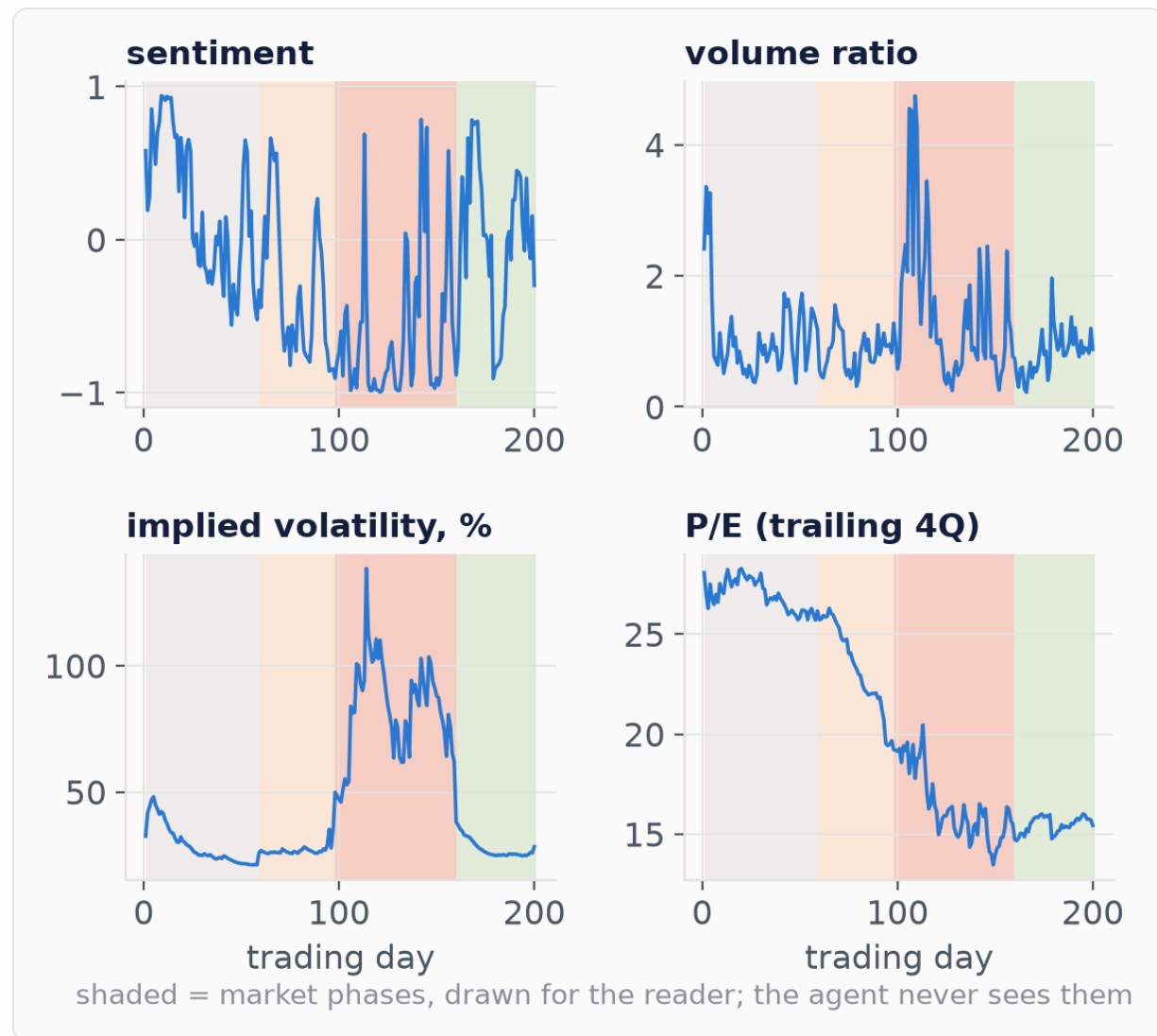
Why the phase cannot be read from the calendar; severity now matters; topped and un-topped bubbles are two reported populations



Observables follow the market, not the label

- Sentiment: autoregressive, driven by current mispricing and recent returns
- Volume: rises with $|\text{return}|$ and $|\text{mispricing}|$
- Implied volatility: 21-day GARCH forecast plus a premium that depends on the volatility state
- Technicals: SMA 20/50, Wilder RSI, MACD with signal, checked against a reference library

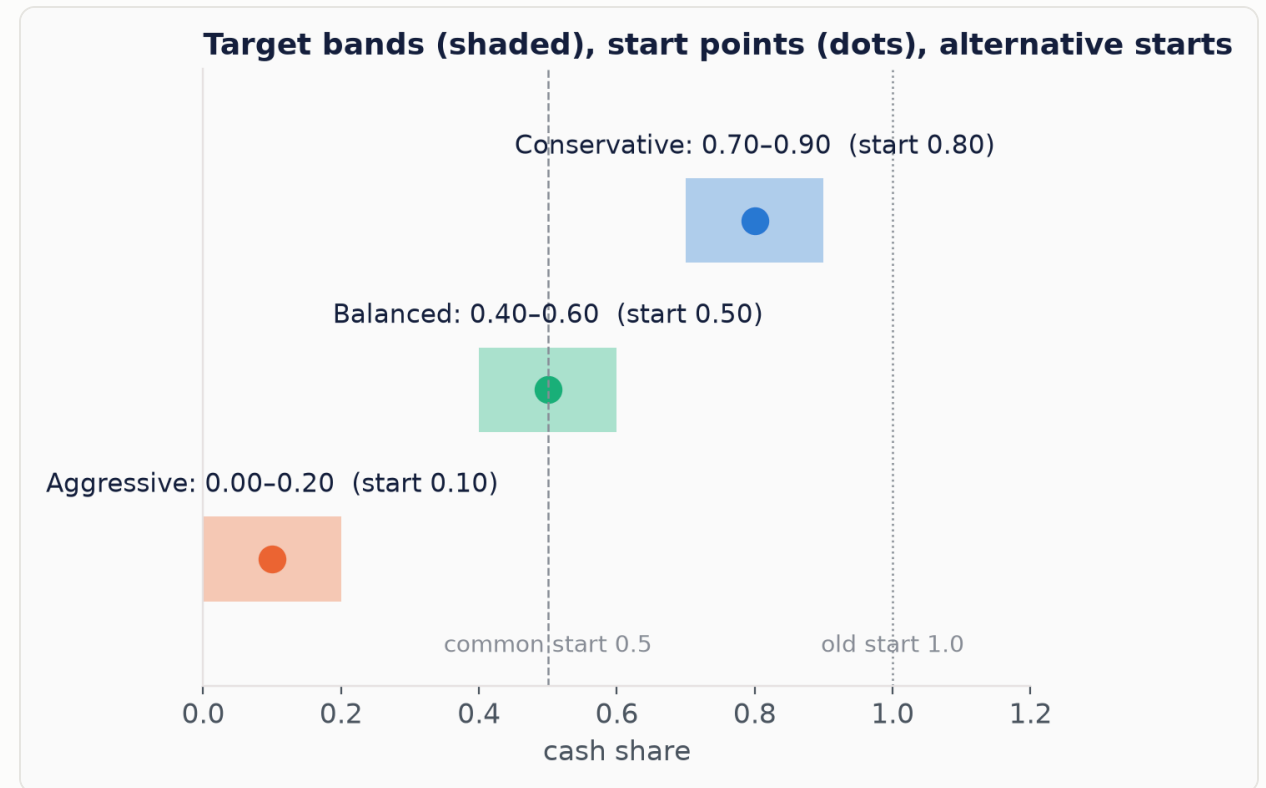
Why no field is computed from the phase label, so none of them is a phase clock



Start allocation and action are design factors

- Start: the persona's band centre (primary), a common 50 % cash, or 100 % cash as a bridge to the old results
- Action: a target cash share; the harness trades the difference, 5 bp per trade, 1-point dead band
- The old BUY / SELL interface is kept behind a flag for the bridge cell

Why the start condition alone could produce the persona effect; now any allocation is reachable in one step



Where the target cash bands come from

Three kinds of source; each is used for one thing. The resulting bands are on the next slide.

1 · Practitioner conventions

set the levels

- Fund companies publish standard conservative, balanced and aggressive portfolios.
- Stocks: about a fifth, about half, almost all.
- As cash: roughly 80 %, 50 % and 10–15 %.

Sources

[Morningstar, US fund category definitions \(2025\)](#)
[Vanguard LifeStrategy funds](#)
[Fidelity model portfolios, methodology](#)
[Betterment, asset-allocation methodology](#)

2 · Portfolio theory

rules out 100 % cash

- Theory gives the stock share from how much risk a person can bear.
- Even the most cautious investor holds some stocks (about 15–25 %).
- No level of caution leads to 100 % cash.

Sources

[Damodaran, equity risk premium data \(2025\)](#)
[Vanguard, "A framework for allocating to cash" \(2024\)](#)

3 · Personality finance & surveys

give the ordering, not the levels

- Personality moves the stock share by only a few points: it fixes the order, not the levels.
- Surveys: 25–40 points between the most and least cautious investors.
- Used to check the ordering and report the spread.

Sources

[Jiang, Peng & Yan \(2024\), Journal of Financial Economics](#)
[Gilliam, Chatterjee & Grable \(2010\), J. Financial Counseling & Planning](#)
[Fieberg et al. \(2025\), CESifo Working Paper 11666](#)

The target cash bands, old and new

The cash share each persona is expected to hold: the single value the old environment stipulated, and the band the new one scores against.

Persona	Old cash target	New cash band (centre)	Comes from
Conservative	1.00	0.70 – 0.90 (0.80)	$\approx 20\%$ equity conventions; risk aversion $\gamma \approx 8\text{--}10$
Balanced	0.50	0.40 – 0.60 (0.50)	50/50 to 60/40 mixes; $\gamma \approx 3\text{--}4$
Aggressive	0.20	0.00 – 0.20 (0.10)	$\geq 85\%$ equity; $\gamma \leq 2$

- Scoring uses the band: distance outside it is the error; the centre is kept for the point version of the score
- Balanced is unchanged; conservative moves from 100 % cash to 80 %; aggressive from 20 % to 10 %
- 100 % cash is kept only as an explicitly labelled ‘fully liquid’ condition in the stated-target track

The prompt equals the table

- The observation table in the paper is generated from the code: 19 fields, all of them rendered in the prompt
- Portfolio state is included; horizon disclosure and field order are switchable factors
- Every output row carries a hash of the generator configuration and of the prompt
- Separate random streams per component, so a seed reproduces exactly across threads

Why paper, prompt and code cannot drift apart, and every run is traceable to its exact inputs

WHAT THE MODEL SEES: day 1 of a crash run (fields in the order they are rendered)

MARKET OBSERVATION:

- Date: Day-1
- Price: 119.26
- SMA20: 112.84
- SMA50: 111.22
- Trend strength (SMA20 vs SMA50, %): 1.46
- Trend regime (-1/0/1): 0
- RSI(14): 74.2
- MACD: 1.2194
- MACD signal: 0.7924
- Volume: 2722311
- Volume ratio (vs 20-day avg): 2.41
- News sentiment (-1..1): 0.58
- Sentiment 5-day avg: 0.52
- Sentiment change vs 5-day avg: 0.06
- Implied volatility (%): 32.5
- P/E (trailing 4Q): 28.1
- Dividend yield (%): 1.27
- Analyst fair-value estimate (\$): 91.56
- Days since last earnings report: 29

PORTFOLIO:

- Cash: \$8000.00
- Holdings value: \$2000.00
- Cash share: 80.00%

Control arms in the harness

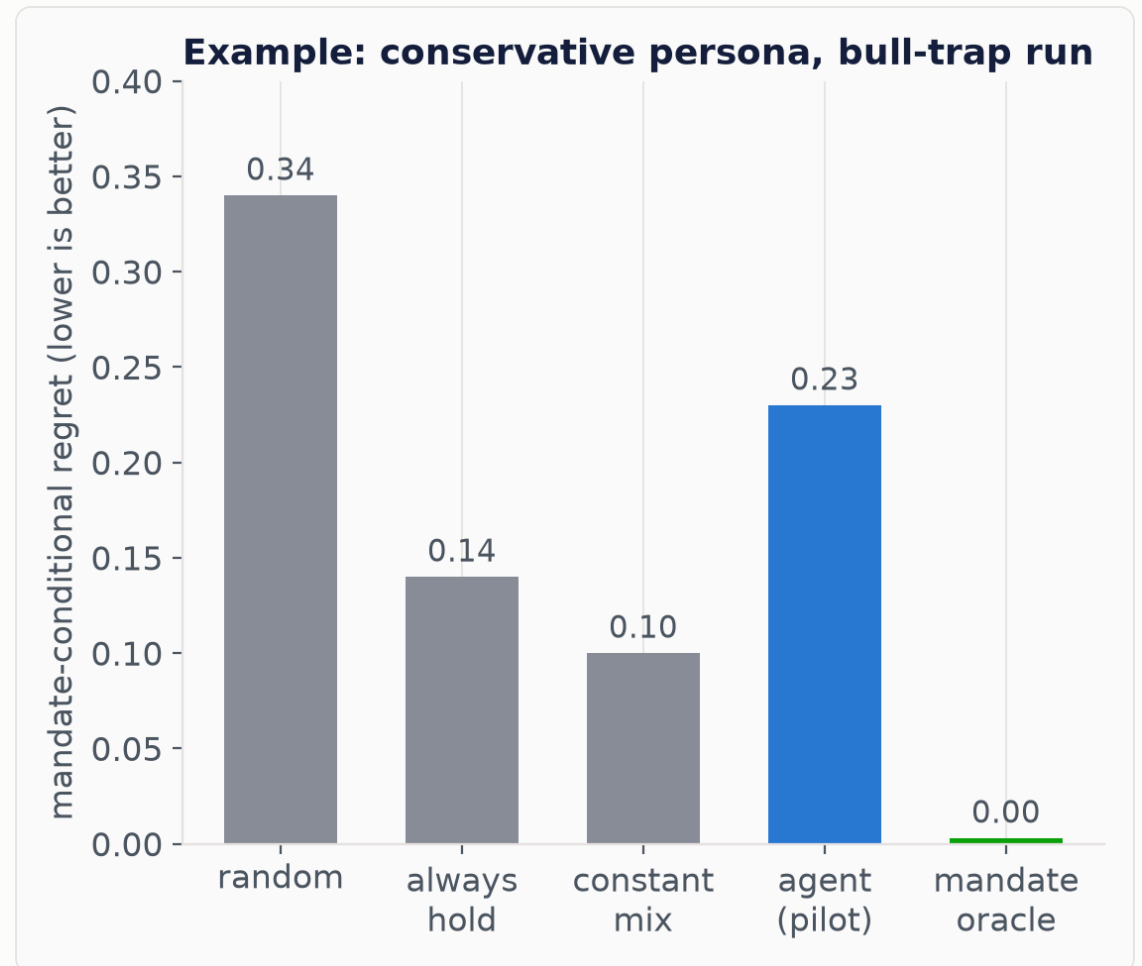
Arm	Persona text	Per-step block	Mandate content
static	yes	none	—
mandate re-injected	yes	ACTIVE MEMORY REFRESH + mandate	own
declarative placebo	yes	neutral boilerplate, same length	—
directive placebo	yes	matched imperative, irrelevant content	—
wrapper only	yes	ACTIVE MEMORY REFRESH, no mandate	—
swapped mandate	yes	ACTIVE MEMORY REFRESH + mandate	another persona's
no-mandate trader	no	none	—
stateful (rolling / full / summary)	yes	prior turns kept as context	own, in the system prompt

- Also varied: wording level, stated-target track, cost visibility, decode replicates, a forked restatement probe
- Stateful arms log context length and the distance since the mandate was last seen

Why this separates the content of the mandate from the force of a reminder, and tests whether context length itself matters

- 11 rule-based baselines on the same path and start: always hold / buy / sell, random, buy-and-hold, constant mix, momentum, mean reversion, value oracle, mandate-conditional oracle, observables oracle
- Steps with |mispricing| below 5 % are not scored; there is no right answer there
- New score, mandate-conditional regret: distance to the best cash share inside the persona's band given the true mispricing

Why the old rationality score rated buy-and-hold near perfect (the floor sat above the ceiling); every number now has a reference point



Additions beyond fixing shortcomings

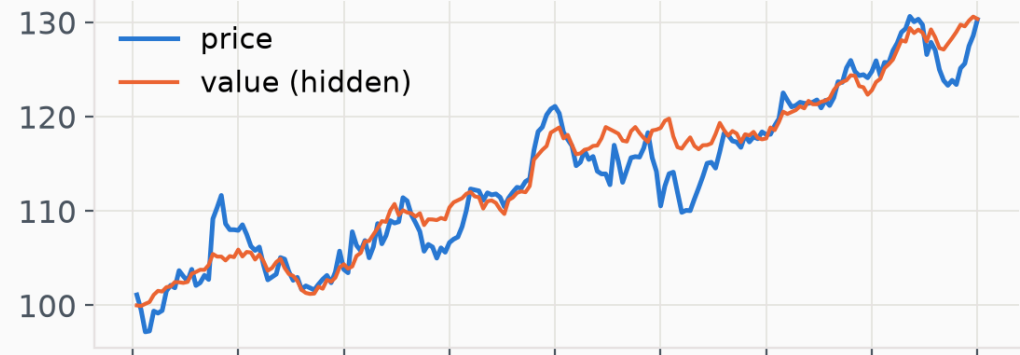
- 1 **A fourth regime, sustained bull:** rising value with no mispricing, a control for 'is it a bubble or just a bull market?'
- 2 **Control arms:** two placebos, wrapper-only, swapped mandate, and a trader with no mandate
- 3 **Stateful arms:** rolling window, full transcript, and summary memory, with context length logged
- 4 **Multi-asset-capable generator:** a three-asset configuration for allocation across holdings
- 5 **Design factors:** trading costs (5 bp, visible or hidden), horizon disclosure, and field order
- 6 **Restatement probe:** 'repeat your mandate' asked without trading, to separate memory from behaviour
- 7 **Baselines and an oracle inside the environment,** plus a regret score with floors and ceilings
- 8 **A statistics track:** model-level contrasts, mixed-effects models, multiple-testing correction
- 9 **Provenance:** a configuration and prompt hash on every output row; the observation table is generated from code

A fourth regime: the sustained bull

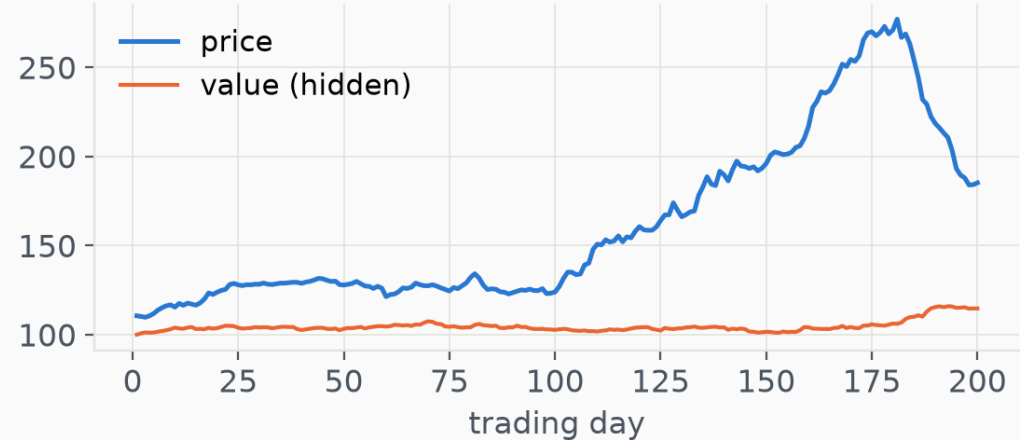
- Value drifts up 0.15–0.25 % a day; there is no scripted mispricing, so price tracks value
- From the price alone it looks like the start of a bubble, but there is nothing to sell
- Rejection sampling keeps only paths that stay inside the control's validity band; the rejection rate is published

Why it separates 'bubble' from 'bull market': an agent that follows its mandate should stay invested here and step back in the bull trap

Sustained bull: price tracks a rising value (nothing to sell)



Bull trap: price leaves value and comes back (a bubble)



Stateful arms and the restatement probe

- Three ways of remembering; the mandate sits in the system prompt in all of them
- Restatement probe: a forked call asks 'state your mandate' every k steps; the answer is logged and the trading trajectory is untouched
- Each stateful arm has a stateless twin on the same seed, so memory is the only difference

Why it tells apart remembering the mandate from being reminded of it, and shows whether context length itself changes behaviour

What the model carries into turn 13

Rolling

last 20 turns verbatim (control arm: last 5)



Full transcript

every turn so far; context grows all run



Summary

a neutral note-taker rewrites the summary every 10 turns; last 5 turns kept raw



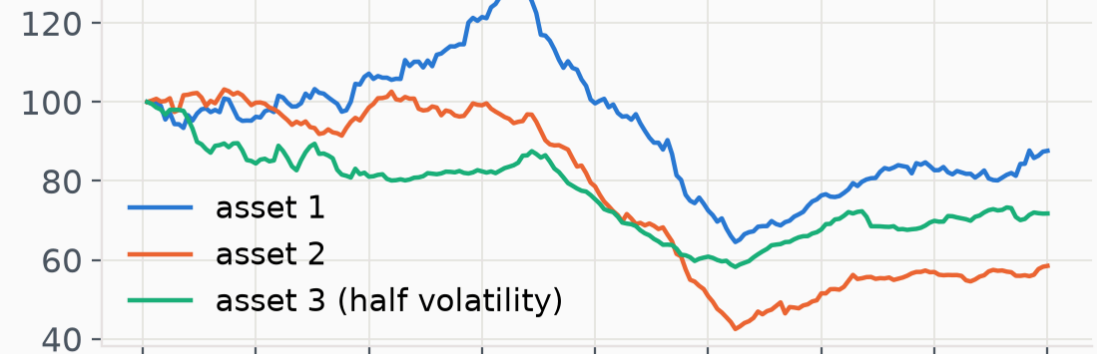
shaded = in context · logged every step: context tokens, turns since the mandate was last seen, the summary text and whether it mentions the mandate

A multi-asset-capable generator

- Three assets: two at full volatility, one at half; a common factor ($\rho = 0.3$) links their shocks
- Each asset has its own value and mispricing; the agent sets a cash share and weights across assets
- Oracle: cash follows 'is any asset cheap or dear?'; weights are scored by distance to the best allocation
- Leakage audits run on the full asset vector, using an asset's own block as the control

Why it moves the test from 'how much cash' to 'which holding' without changing the rules of the game

Three assets, one crash run: prices (day 1 = 100)



Each asset carries its own mispricing, %



Design factors the experiment can switch

Factor	Levels	What it tests
Trading cost	0 or 5 bp per trade; shown or hidden in the prompt	does a visible cost change churn?
Horizon disclosure	length of the run told / not told	end-of-run behaviour
Field order	canonical / shuffled	position effects inside the prompt
Start allocation	band centre · common 50 % · 100 % cash	the start-condition confound
Wording level	original · rewritten · defensive	is it the words or the mandate?
Track	A: target stated in the prompt · B: persona text only	instruction following vs revealed preference
Decode replicates	temperature 0.2, n repeats per cell	sampling noise around every estimate

- Every factor is a column in the output, so any cell can be re-run or re-scored on its own

Why effects that used to be baked into the environment are now things the experiment can switch on and off

Question	Method
Does the arm change the allocation?	model-level contrasts with Cliff's δ and Hedges g ; Benjamini–Hochberg correction across models
Is it the persona, or just that run?	mixed-effects model with a random intercept per run; the static arm is the reference
Is it time, or the market phase?	25-day window means with phase and time in one model; circular-shift null for phase-free runs
Does memory matter beyond the phase?	paired sign-flip test: stateful minus stateless on the same seed
Is the persona even there on day 1?	separability gate (ordering, effect size, band-hit rate) before any drift is scored

- Reports and tables are generated from the run files, so the numbers in the paper come from code

Why every headline number now has a null, an effect size and a multiplicity correction behind it

Stylized-facts checklist, 50 seeds per scenario

Property	Before	After	Target	Result
No autocorrelation in calm returns	ACF 0.16	ACF 0.08	< 0.15	PASS
Fat tails (excess kurtosis > 1.5)	4 % of seeds	79 %	80 %	margin
Volatility clustering (Ljung–Box)	39 %	60 %	80 %	FAIL
Mispricing half-life	0 days	72 days	≥ 60 days	PASS
Drawdown spread, discount 0.55 vs 0.85	0.5 pp	18 pp	20 pp	near
Bubble: topped share / peak P/V	—	44 % / 2.2	40–60 % / 1.6–2.5	PASS
Phase guessable from the calendar day	100 %	65 %	< 80 %	PASS
Sentiment persistence / link to returns	0.6 / 0.08	0.87 / 0.37	0.7–0.9 / 0.25–0.55	PASS
Implied vol in panic / corr. with realised	34 % / 0.39	59 % / 0.39	60–100 % / 0.4–0.8	margin

8

pass (before: 3)

7

fail (before: 10): 4 within sampling error, 2 structural, 1 calibration

7

not applicable

nothing re-tuned to pass

thresholds were fixed before the runs; every miss is reported

Question	Before	After
Can a formula on the shown fields reproduce the hidden value?	yes, within 0.2 %	no, ≥ 12 % off
Do the non-price fields reveal the phase beyond what price tells?	+5 pp	+6.6 pp (limit 10 pp)
Share of flat-market steps with a right answer ($ x \geq 5$ %)	0 %	72 %
Is mispricing predictable from price history? (R^2 , calm)	0.90	0.83

- The value is no longer recoverable from any field, and the fields do not leak the phase
- Mispricing stays inferable from price history by design (smooth value, persistent mispricing); that is what makes a fundamentalist mandate playable, and it is stated openly

Error of the best formula that tries to rebuild the hidden value from the shown fields (bigger is better: the value is harder to back out)



Flat-market steps that have a right answer (bigger is better: the flat market now tests decisions)

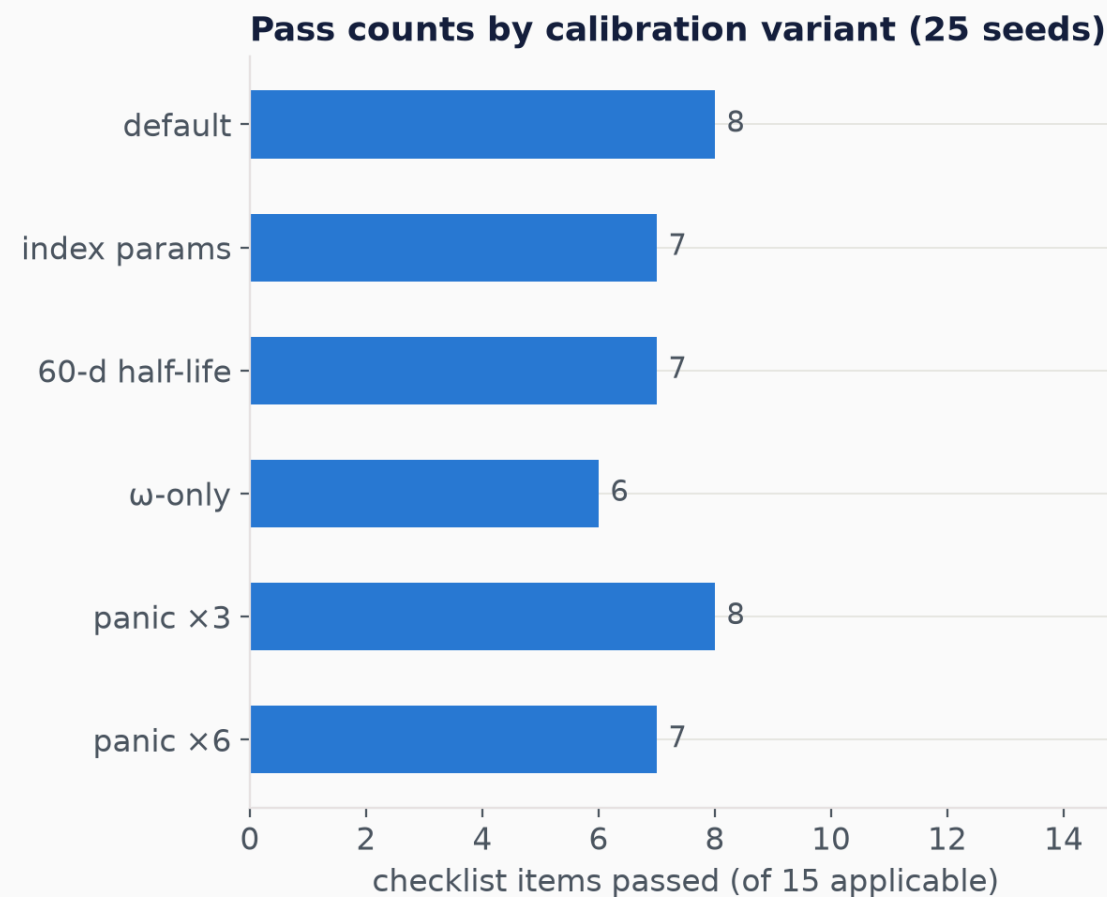


Phase information carried by the non-price fields, beyond price (limit 10 pp; both versions stay under it)



- A policy fitted only on what the agent sees comes within 0.02–0.08 regret of the true-value oracle → the environment withholds the **level** of value, not the **direction** of mispricing
- Checklist pass counts are stable across the main calibration variants (25 seeds each):

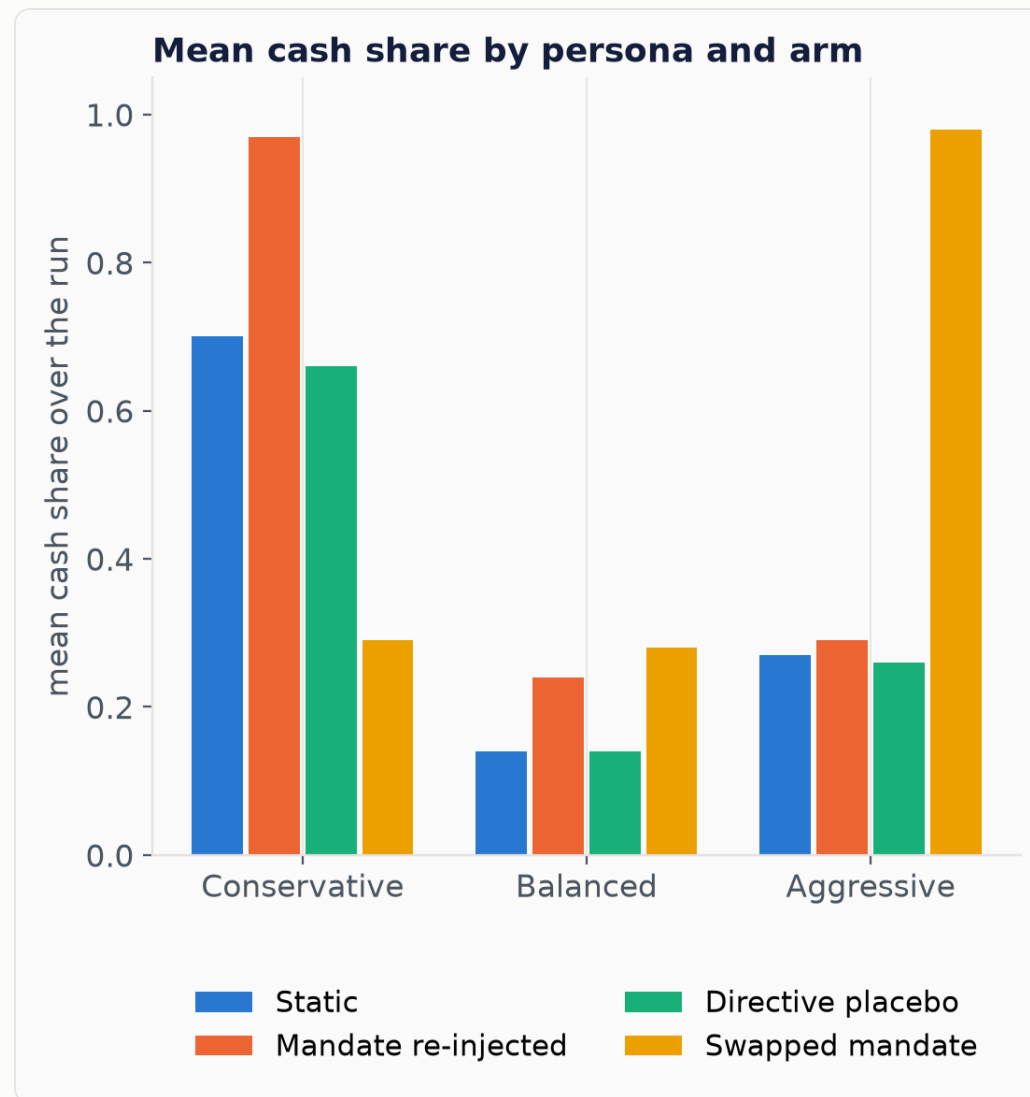
Variant	Pass / fail
Default	8 / 7
Published index parameters	7 / 6
60-day half-life	7 / 8
Variance multipliers on ω only	6 / 7
Panic variance $\times 3$ / $\times 6$	8 / 5 · 7 / 6



Pilot: Gemini 2.5 Flash, one seed, 200 days, 48 runs

- 100 % of calls parsed, no fallbacks; restatement probes coherent; the stateful arm reaches $\approx 20k$ tokens by day 200
- Numbers are the mean cash share over the run; one model, one seed: a pipeline check, not a result
- The swapped mandate moves the allocation to the injected content; the directive placebo tracks the static arm

Persona (start)	Static	Mandate	Placebo	Swapped
Conservative (0.80)	0.70	0.97	0.66	0.29
Balanced (0.50)	0.14	0.24	0.14	0.28
Aggressive (0.10)	0.27	0.29	0.26	0.98



Summary

- ✓ Ground truth is explicit; valuation-field leakage is fixed; phases are randomised; volatility is real
- ✓ Target cash bands are grounded in practitioner conventions and portfolio theory, with sources
- ✓ Start allocation, action, costs and control arms are design factors, not accidents
- ✓ Added: a sustained-bull control regime, stateful arms with a restatement probe, a multi-asset generator, a statistics track
- ✓ Baselines, a non-degenerate score and pre-registered checks; every failure is reported
- ✓ The pilot exercises every piece end to end